

Benchmarking of NQCH's quantum computer

April 30, 2026

1. Report of Changes

Platform: sinq20

Calibration-id: *ec305fcaf377a05f4fb391a8ee18282bfae18df6*

Calibration date: 2026-04-28 07:58:51

Calibration note: chore(sinq20): update readout parameters

Experiment-id: 20260429212203

Experiment date: 2026-04-29 21:22:04

Experiment note: *temporary note!!!*

Platform: sinq20

Calibration-id: *4dc4082f38a53222b3956c22202d32a520d4bc78*

Calibration date: 2026-01-15 02:03:03

Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538

Experiment date: 2026-01-18 16:15:38

Experiment note: *temporary note!!!*

2. Version Comparison

Library	Version	Library	Version
qibo	0.2.23	numpy	2.4.4
qibolab	0.2.9	qibocal	0.2.5
matplotlib	3.10.8	scipy	1.17.1
scikit-learn	1.8.0	pandas	2.3.3
networkx	3.6.1	sympy	1.14.0
torch	2.11.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



8. Mermin

Runtime	Qubits	Max
-	-	-



Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in [[11, 6]].

Runtime	Qubits	Max prob
6.10 sec	11, 6	0.86



Runtime	Qubits	Max prob
2.36 sec	2, 3	0.89

